Package 'gltbnn'

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Type Package

Title Variational Bayesian Neural Networks with Grouped Shrinkage Priors

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Description Fit Bayesian neural networks for Gaussian regression with global, grouped horseshoe, or global-local-tail shrinkage on the incoming weights and bias of each hidden unit. The global-local-tail model learns a separate tail shape for each hidden layer. Variational inference yields draws of the regression function and posterior predictive draws for new observations. The tail-adaptive prior builds on Lee et al. (2024) <doi:10.1214/24-EJS2317>.

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Imports stats

URL https://yain22.github.io/Bayesian_NN_GLT/, https://github.com/yain22/Bayesian_NN_GLT

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`gltbnn_fit` *Fit a Bayesian neural network with grouped shrinkage priors*

Description

Fit a fully connected neural network for a continuous response. Each hidden layer uses a rectified linear unit (ReLU) activation. The final layer computes an affine function and the response has Gaussian observation noise.

Usage

```
gltbnn_fit(x, y, hidden = c(10L, 10L), prior = c("GLT", "GL", "G"),
           control = list(), seed = 1L, noise_prior = NULL)

## S3 method for class 'gltbnn_fit'
print(x, ...)
```

Arguments

`x`	For fitting, a numeric matrix or numeric data frame with one row per observation and one column per predictor. Supply at least two observations and one predictor. For printing, a fitted `gltbnn_fit` object.
`y`	Numeric vector of responses, one per row of `x`. All training variables must be finite and have positive sample standard deviations.
`hidden`	Positive integer widths in input-to-output order. For example, `c(4L, 8L)` creates four neurons in the first hidden layer and eight in the second. Biases are included automatically.
`prior`	One of `"G"` (global), `"GL"` (grouped horseshoe), or `"GLT"` (global-local-tail, the default).
`control`	Named list of overrides for the fitting controls described below.
`seed`	Nonnegative integer random seed. The previous `.Random.seed` is restored after fitting.
`noise_prior`	Prior for the observation variance on the standardized response scale. The default `NULL` uses the paper's Jeffreys prior. For a proper inverse-gamma prior, supply a numeric vector named `shape` and `scale`, both finite and strictly positive. For example, `c(shape = 3, scale = 2)` has mean 1 in standardized units.
`...`	Additional arguments to the print method, currently unused.

Details

The incoming weights and bias of neuron k in hidden layer l form a group of $d_l = K_{l-1} + 1$ coefficients. Here K_l is the width of layer l and K_0 is the number of predictors. Conditional on the scales, these coefficients have independent zero-centered Gaussian priors with standard deviation $\sigma\lambda_{kl}$. The quantity σ^2 is the Gaussian observation noise variance on the standardized training scale. The same σ sets the units for the hidden weight prior. The default uses the improper density kernel $1/\sigma^2$. Its posterior is proper almost surely when the responses have a joint density conditional on the predictors. Some nonconstant data can yield an improper posterior. For example, under G with one hidden layer, three or more distinct univariate inputs and exactly affine responses with nonzero

slope can have this property. Interpret variational draws as approximate posterior draws when the posterior is proper.

The alternative inverse-gamma prior gives a proper posterior for every finite dataset accepted by this function. For variance s, shape a and scale b, its density is proportional to $s^{-(a+1)} \exp(-b/s)$. The prior uses standardized fitting units. Its scale parameter on the original response scale equals b times the training sample variance of `y`.

With `"G"`, every neuron in a layer shares the scale $\lambda_{kl} = \tau_l$. With `"GL"`, $\lambda_{kl} = \tau_l r_{kl}$, where each unit-specific multiplier has an independent standard half-Cauchy prior. Both priors give τ_l a half-Cauchy prior with scale $1/K_l$. With `"GLT"`, the unit scales follow a generalized Pareto distribution conditional on the layer scale τ_l and tail shape ξ_l. The model learns both for each layer. Given ξ_l, the prior on τ_l is inverse-gamma with shape $1 + K_l/\xi_l$ and scale 1. The prior on $\log \xi_l$ is standard normal truncated below at $-\log 2$.

All three models keep the chosen hidden widths and use a separate scale for the final affine layer. Its weights and intercept have independent zero-centered Gaussian priors conditional on a shared standard deviation κ, which has a half-Cauchy prior with scale 5.

The function centers each training variable and divides by its sample standard deviation. It then initializes the network and fits a variational approximation with independent Gaussian weight factors. The entries below are available in `control`.

`init_steps` Number of deterministic-network optimizer steps used to initialize the weight means (default 600).

`outer`, `weight_steps` Number of outer iterations (default 60) and weight updates per iteration (default 10).

`lr`, `grad_clip` Initial learning rate for each variational weight update (default 0.003) and gradient norm cap (default 10).

`max_backtracks` Maximum number of learning-rate halvings per weight update (default 10).

`logsd_bounds` Lower and upper bounds on the log standard deviations of the variational weight factors (default `c(-15, 3)`). At both endpoints, `exp(2 * bound)` must give a positive finite variance in double precision. Fitting starts at `log(0.01)` or at the nearest bound if that value falls outside the interval.

`weight_draws`, `noise_draws` Monte Carlo draws per weight update (default 4) and per noise update (default 16).

`monitor_draws`, `endpoint_draws` Monte Carlo draws for the evidence lower bound (ELBO) recorded during training (default 64) and at the independently sampled endpoint (default 256).

With the Jeffreys prior, the objective uses the unnormalized joint density with the arbitrary additive constant of the log prior set to zero. Under this convention the endpoint value is a diagnostic for the variational fit. Normalized model evidence requires a proper prior. The inverse-gamma option includes the normalized log prior in the objective. Its component `noise_neg_kl` combines the expected log prior and factor entropy as minus the KL divergence from the fitted variance factor to its prior. The Jeffreys fit records `noise_prior` and `noise_entropy` separately. Each of the four training draw counts must be an even integer of at least two because the estimator pairs each draw with its negative. With just one pair, the Monte Carlo standard error is undefined. Training uses the requested iteration budget.

Value

An object of class `gltbnn_fit`. Fields include `prior`, `hidden`, the training dimensions `n` and `p`, `predictor_names`, the training centers and scales in `preprocess`, `control`, the chosen observation-variance prior in `noise_prior`, and `seed`. The `engine` list contains the variational weight parameters (`parameters`), scale factors (`state`), iteration history (`trace`), and the final Monte Carlo

ELBO estimate (`endpoint$value`). Use `gltbnn_shrinkage` for unit and layer summaries in the paper's notation. In the stored factors, `engine$state$g[[l]]$tau` represents τ_l^2 under G and GL, and τ_l under GLT. For GLT, `g[[l]]$tail$r` is the variational expectation of $1/\xi_l$ and `g[[l]]$tail$eta_mean` is the expectation of $\log \xi_l$. Inverse-gamma factors store their shape in `a` and their scale in `b`. The reciprocal has a gamma distribution with shape `a` and rate `b`. The print method returns its input invisibly.

See Also

`gltbnn_shrinkage`, `predict.gltbnn_fit`

Examples

```
x <- data.frame(z = seq(-1, 1, length.out = 21))
y <- x$z^2 + 0.1 * cos(9 * x$z)
# A short run to illustrate the interface.
fit <- gltbnn_fit(x, y, hidden = 3L, prior = "GL",
                  control = list(init_steps = 10L, outer = 2L, weight_steps = 2L))
fit
new_x <- data.frame(z = c(-0.5, 0, 0.5))
predict(fit, new_x)
predict(fit, new_x, type = "prediction_interval", level = 0.9)
```

`gltbnn_shrinkage` *Inspect shrinkage for each hidden unit*

Description

Summarize the fitted scale shared by the incoming weights and bias of each hidden unit. A smaller typical conditional prior standard deviation means these coefficients are more tightly concentrated around zero under their conditional prior.

Usage

```
gltbnn_shrinkage(fit)
```

Arguments

`fit`	A model returned by `gltbnn_fit`.

Details

Conditional on the scale and on the observation-noise variance σ^2, each incoming coefficient has a zero-mean Gaussian prior with variance $\sigma^2\lambda_{kl}^2$. The fit assigns the observation-noise variance its own variational factor, independent of the scale factors under q. The expected conditional prior precision is therefore $E_q[\sigma^{-2}]E_q[\lambda_{kl}^{-2}]$. The geometric prior standard deviation uses the expected log standard deviation. The variational posterior standard deviation of the weight itself comes from its separate Gaussian weight factor.

Under G all units in a layer share one scale and have identical summaries. Under GL the unit scale is the product of the layer and local scales. Under GLT the unit scales follow a generalized Pareto distribution given the layer scale and tail shape. The inverse-gamma scale factor is fitted to τ_l^2 under G and GL and to τ_l under GLT. The reported layer scale uses τ_l itself in all three models.

Value

A data frame with one row per hidden unit, ordered by layer and unit.

`layer`, `unit` One-based indices of the hidden layer l and the receiving unit k within that layer.

`group_size` Number of coefficients entering the unit, including its bias.

`scale_precision` Expected inverse squared unit scale $E_q[\lambda_{kl}^{-2}]$ under the fitted variational distribution q. This excludes the observation-noise factor.

`geometric_prior_sd` Geometric mean under q of the conditional prior standard deviation of an incoming coefficient, $\exp\{E_q[\log(\sigma\lambda_{kl})]\}$, on the standardized fitting scale.

`geometric_layer_scale` Geometric mean of the layer scale, $\exp\{E_q[\log\tau_l]\}$.

`geometric_tail_shape` For GLT, the geometric mean of the layer's generalized Pareto tail shape, $\exp\{E_q[\log\xi_l]\}$. G and GL have no fitted tail shape and return `NA_real_`.

Both layer summaries repeat on every row from the same layer.

See Also

`gltbnn_fit`

Examples

```
x <- data.frame(z = seq(-1, 1, length.out = 15))
y <- x$z^2
fit <- gltbnn_fit(x, y, hidden = 3L, prior = "GLT",
                  control = list(init_steps = 10L, outer = 2L, weight_steps = 2L))
units <- gltbnn_shrinkage(fit)
units
unique(units[c("layer", "geometric_layer_scale", "geometric_tail_shape")])
```

`predict.gltbnn_fit` *Predict conditional means or future responses*

Description

Use draws from a fitted network's variational posterior to predict at new observations. Predictions are returned in the original response units.

Usage

```
## S3 method for class 'gltbnn_fit'
predict(object, newdata,
       type = c("mean", "mean_draws", "response_draws", "mean_interval",
                "prediction_interval"),
       draws = 256L, level = 0.95, seed = 1L, ...)
```

Arguments

object	A model from gltbnn_fit.
newdata	Numeric matrix or numeric data frame with one row per new observation. Supply the same predictors as in training. When both data sets have column names, predictors are matched by name. Otherwise supply them in training order.
type	"mean" averages draws of the conditional mean. "mean_draws" returns those draws. "response_draws" samples future responses including Gaussian observation noise. "mean_interval" and "prediction_interval" return equal-tailed pointwise intervals for the conditional mean and a future response, respectively.
draws	Positive integer number of posterior draws. At least two are required for intervals.
level	Probability within each pointwise interval, strictly between zero and one.
seed	Nonnegative integer random seed. The previous .Random.seed is restored after prediction.
...	Reserved for future use.

Details

In each draw the same sampled network evaluates all rows of newdata. A mean draw is its network output. For a response draw, the method also samples a noise variance from its variational factor and adds independent Gaussian errors at the new observations, conditional on that variance. Intervals use sample quantiles of the corresponding draws.

Value

For "mean", a numeric vector with one entry per observation. For draws, a numeric matrix with one row per observation and one column per draw. For intervals, a two-column matrix named lower and upper.

See Also

gltbnn_fit for a fitting and prediction example.

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